

Program 7

Deploy a web Application to kubernetes using Minikube

Program Structure:

program7/

----->nginx-deployment.yaml

Procedure:

step 1: Start the minikube

minikube start

Ccheck the status of minikube

minikube status

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program7$ nano nginx-deployment.yaml
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program7$ minikube start
minikube v1.37.0 on Ubuntu 22.04
Using the docker driver based on existing profile
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.48 ...
Restarting existing docker container for "minikube" ...
Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
Verifying Kubernetes components...
  ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
  ■ Using image docker.io/kubernetesui/dashboard:v2.7.0
  ■ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

☀ Enabled addons: storage-provisioner, default-storageclass, dashboard

! /usr/bin/kubectl is version 1.30.14, which may have incompatibilities with Kubernetes 1.34.0.
  ■ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
🏠 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program7$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program7$
```

step 2: Create nginx-deployment.yaml file and add the content

nano nginx-deployment.yaml

```

GNU nano 6.2 nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80

```

step 3 : Apply the nginx-deployment.yaml file
 docker apply -f nginx- deployment.yaml
 Check status
 docker get pods
 docker get deployments

```

1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/mini$ nano nginx-deployment.yaml
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/mini$ kubectl expose deployment nginx-deployment \
> --type=NodePort \
> --port=80
service/nginx-deployment exposed
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/mini$ kubectl get services
NAME          TYPE        CLUSTER-IP   EXTERNAL-IP  PORT(S)    AGE
kubernetes    ClusterIP   10.96.0.1    <none>       443/TCP    11m
nginx-deployment NodePort    10.98.106.77 <none>       80:31274/TCP 29s
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/mini$ minikube services nginx-deployments
Error: unknown command "services" for "minikube"

Did you mean this?
  service

```

step 4: Expose the deployment as service
 kubect expose deployment nginx-deployment --type=NodePort --port=80
 kubectl get service
 minikube service nginx-deployment

```

1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/mini$ kubectl get svc
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
kubernetes           ClusterIP   10.96.0.1     <none>         443/TCP          16m
nginx-deployment     NodePort    10.98.106.77  <none>         80:31274/TCP     5m6s
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/mini$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    2/2     2            2           14m
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/mini$ kubectl expose deployment nginx-deployment \
Error from server (AlreadyExists): services "nginx-deployment" already exists
Error from server (NotFound): deployments.apps " " not found
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/mini$ minikube service nginx-service
X Exiting due to SVC_NOT_FOUND: Service 'nginx-service' was not found in 'default' namespace.
You may select another namespace by using 'minikube service nginx-service -n <namespace>'. Or list out all the services using 'minikube service list'

1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/mini$ minikube service nginx-deployment

```

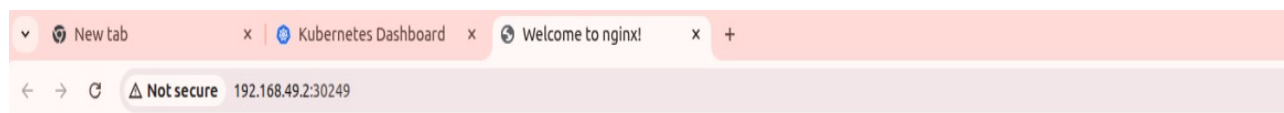
NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:31274

```

🌐 Opening service default/nginx-deployment in default browser...
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/mini$ minikube service nginx-deployment
Opening in existing browser session.

```

output



Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.